

Soham Roy

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EDUCATION

University of Engineering and Management, Jaipur	2024 – Present
<i>B.Tech in Computer Science Engineering</i>	CGPA: 9.14/10 (S1: 9.38 S2: 8.73 S3: 9.30)
Class 12 (Higher Secondary)	2024
	GPA: 8.64
Class 10 (Secondary)	2022
	GPA: 9.89

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C
ML & Data Science: Scikit-learn, LightGBM, XGBoost, Pandas, NumPy, Matplotlib
Modeling: Classification, Regression, Ensemble Methods, Quantile Regression, Spatial CV
Backend & Deployment: FastAPI, Node.js, Express.js, React.js
Tools: Git, Arduino, ESP32
Domains: Quantitative Finance, Geospatial Analysis, Astrophysics Data, IoT

PROJECTS

Stellar Analytics <i>Scikit-learn, LightGBM, FastAPI, React</i> Demo Code Winner – IIT BHU Technex '26	2026
– Benchmarked 16 ML models on NASA Kepler data (9,564 records, 50 features) for exoplanet classification and radius prediction; best classifier achieved ROC-AUC 0.9504.	
– Built FastAPI-powered inference pipeline with interactive visual diagnostics and feature importance analysis.	
AlgoZen – Quantitative Trading System <i>Python, Pandas, NumPy</i> Code	2026
– Built a regime-aware NIFTY 500 systematic trading strategy with walk-forward backtesting achieving 10.92% CAGR and Sharpe 1.129.	
– Engineered ATR trailing stops, volatility targeting, and circuit breakers using a custom Python backtesting engine.	
ShadowMap – Urban Heat Island Prediction <i>XGBoost, FastAPI, React</i> Demo Code	2025
– Built a geospatial ML system predicting urban surface temperature using Sentinel-3, Landsat NDVI, and OSM features; achieved R^2 0.926.	
– Integrated quantile regression and intervention simulation into an interactive Leaflet.js choropleth interface.	
Carbon-Wise <i>React, Node.js, Express.js, SQLite</i> Code	2025
– Built a vehicle lifecycle emissions platform comparing EV vs ICE carbon output across manufacturing, operation, and disposal phases.	

EXPERIENCE & LEADERSHIP

Core Organizer – ACEHack 5.0	2026
<i>Rajasthan's Premier Student-Run Hackathon</i>	
– Led operations for one of Rajasthan's largest student hackathons with 10,000+ registrations and 500+ on-site participants.	
– Managed sponsor logistics, judge onboarding, outreach coordination, and real-time event execution.	
Web Developer – International Academic Conferences	2025 – 2026
<i>UEM Jaipur</i>	
– Designed and deployed conference websites for ICNIC 2026 and ICCASA 2026 , building custom frontends alongside WordPress CMS.	

ACHIEVEMENTS

Winner – Stellar Analytics Competition, Technex '26, IIT BHU
Top 3 – Case Study Competition, Technex '25, IIT BHU
Round 5 Qualifier – National Hackathon, MNIT Jaipur
Hackathon Finalist – Competed in 10+ national hackathons; finalist in 5 major events.
Autonomous Drone Build – UDAAN Workshop | [Post](#)